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Why AI-Generated Image Detectors Fail on Modern Diffusion Generators: A Diagnostic Study of Cross-Architecture Generalization and Failure Modes

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Abstract

This is a diagnostic study: it trains nothing and proposes no detector, and instead measures how five pretrained, publicly released AI-generated image detectors behave on five generators spanning three architecture families, with every generator verified absent from every detector's training data. Evaluation uses 6,000 real and 6,000 generated images per generator, 18,300 per class for SDXL, and paired subsets of 2,000 for the compression experiments. Detectors disagree about what is hard: one fails worst on FLUX.1 (7.0% fake-accuracy), another on SDXL (54.6%), two others on BigGAN (0.4% and 9.6%), and paired McNemar tests on identical images confirm the gaps with one non-significant exception between two same-lineage detectors. Two detectors trained on identical data but different architectures produce an identical difficulty ordering across all five generators ($\rho = +1.000$, exact $p = 0.017$) while differing 37.6 points in mean accuracy. Testing the ensembling prediction this suggests, a different-lineage partner lifts worst-generator accuracy from 9.6% to 62.9% where a same-lineage partner of different architecture lowers it to 0.3%. Calibration error rises with failure severity rather than falling, reaching 0.503 where fake-accuracy is 0.4%, so these models stay confident and become wrong outside their training distribution. Two measurement hazards bound every absolute number here. A resize convention rarely specified in released code moves the same checkpoint by over 15 points, in opposite directions for two checkpoints trained on identical data. And because these benchmarks pair PNG generated images against JPEG reals, re-encoding only the generated side at JPEG quality 75 takes the best clean-data FLUX.1 detector from AUC 97.1 to 39.1, below chance, while the other four lose between 1 and 13 points.

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Keywords: AI-generated image detection; synthetic image forensics; cross-generator generalization; diffusion models; adversarial training; calibration; reproducibility

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1. Introduction

A detector's accuracy number is only useful if it predicts behavior on data the detector has not seen. A detector trained on one generator's outputs usually loses most of its accuracy on another generator's outputs, and the field has largely explained this with a single story: newer generators produce artifacts that older detectors were not built to catch. That story predicts a pattern. Detectors should fail worst on the newest generators, and should fail in broadly similar ways.

We test the prediction by running five independently trained, publicly released detectors against five generators spanning three architecture families, and it does not hold. The detectors disagree about which generator is hardest: one fails worst on FLUX.1, another on SDXL, and two others on BigGAN, the oldest generator in the set. In the sharpest case, one detector recovers 5,452 FLUX.1 images that another misses while losing only 8 in the opposite direction, on the identical image set. Generator age is not the variable driving detector failure; the detector's own training distribution is.

This is a diagnostic paper: it proposes no detector and trains nothing, and every number comes from inference over released checkpoints and datasets.

Five contributions follow. We report a controlled generalization audit of five detectors on five generators, with training/test overlap verified against primary documentation rather than inferred from the accuracy pattern such a check is meant to validate, and every gap backed by a paired McNemar test (Section 4.2). We decompose the disagreement using two detectors trained on identical data but different architectures (Section 4.3), then test the operational prediction that follows, that detectors should be combined for training-lineage diversity rather than architectural variety (Section 4.4). We give a calibration account of the failure mode, with per-detector reliability diagrams (Section 4.7). Finally we report two measurement hazards that bound how any accuracy on these benchmarks should be read, a resize convention (Section 5) and a compression asymmetry built into the benchmarks themselves (Section 6).

On terminology: "deepfake" originally named the narrower problem of face swapping and reenactment, where the manipulated region is a face inside an otherwise real photograph. We study the broader task of deciding whether an entire image was synthesised, over ImageNet-style object and scene imagery, which is more precisely called AI-generated image detection. Nothing here is a claim about face-manipulation detection.

2. Related Work

Cross-generator generalization. Wang et al. [1] established the standard protocol of training on one GAN family (ProGAN) and testing against unseen GANs. Zhu et al.'s GenImage [2] extended it to diffusion models at million-image scale and is the training source for three of the five detectors we evaluate. Ojha et al.'s UniversalFakeDetect [3] showed that a frozen CLIP embedding with a linear probe, trained only on ProGAN, transfers to GANs and early diffusion models better than end-to-end training; we use it as our CLIP-embedding baseline. Tan et al.'s NPR [4] instead targets neighbouring-pixel artifacts left by a generator's up-sampling operations, and is the one detector here whose training data and detection signal are both unrelated to CLIP. Work that keeps the CLIP embedding and changes the head, such as C2P-CLIP [18], or that detects diffusion images by reconstruction error, such as DIRE [15], shows that the field's usual response to the generalization gap is architectural, rather than asking why a given detector's training distribution predicts where it fails.

The adversarial-training comparison. Zhou et al. [5] attribute cross-generator failure to latent prior bias: a detector can key on artifacts of its training generator's initial-noise sampling rather than of the generation process, a shortcut Xu et al. [14] showed exists to be learned by recovering the random seed of a diffusion model's initial noise from the generated image. Their fix, On-Manifold Adversarial Training (OMAT), fine-tunes against adversarially optimized initial latents of the same training-time generator, and their GenImage++ benchmark adds held-out subsets from FLUX.1 [6] and Stable Diffusion 3 [7], plus the in-the-wild Chameleon collection [16]. We neither modify nor retrain their checkpoints; we evaluate them alongside

unrelated detectors on a wider generator set, and test whether their preprocessing choice is load-bearing, which their evaluation did not need to ask.

Two boundaries of scope. The detectors here are not face-manipulation deepfake detectors, which target a local edit inside a genuine photograph and are benchmarked on corpora such as FaceForensics++ [19] and Celeb-DF [20]; our generators synthesise every pixel, so the two settings do not share an artifact model. And where an interpretability literature attributes failure to spectral artifacts [17, 21] or visualises which regions drive a classifier [22], we take a coarser route in Section 4.7, characterising the predicted probability distribution and its calibration error, which assumes nothing about which regions carry the evidence.

3. Methodology

Every result comes from running a pretrained, publicly released detector checkpoint over a pretrained, publicly released image dataset, with no training and no image generation. This began as a hardware constraint, since the single 8 GB consumer GPU used throughout cannot hold a generation pipeline and a detector at once, but it makes the pipeline reproducible without generation infrastructure.

3.1. Detectors

Table 1 lists the five detectors, spanning 2023 to 2025. Three use a frozen CLIP [11] ViT-L/14 encoder with a linear (or LoRA [12]-adapted) probe; the other two are convolutional, one keying on pixel and upsampling artifacts and one a plain ResNet-50. Training composition was confirmed from each detector's own repository documentation, and for the GenImage++ checkpoints from the source paper (Section 3.4), rather than inferred from how well each performed.

The GenImage++ ResNet-50 was trained on exactly the same data as the GenImage++ CLIP baseline but is a conventional CNN with no CLIP backbone, so it is the one pair here that varies architecture while holding training data fixed. That is what lets Section 4.3 separate what the training distribution does from what the representation does.

Table 1. Detectors evaluated. All five are used as released, with no fine-tuning performed for this study.

Detector	Signal	Training data (verified)	Year	Source
UniversalFakeDetect	CLIP ViT-L/14 embedding + linear probe	ProGAN only, 20 LSUN categories	2023	[3]
NPR	Pixel / upsampling artifact (ResNet-50)	ProGAN only, 4 LSUN categories	2024	[4]
GenImage++ ResNet-50	Plain CNN classifier (no CLIP)	GenImage SDv1.4 subset only	2025	[5]
GenImage++ CLIP	CLIP ViT-L/14 embedding + linear probe	GenImage SDv1.4 subset only	2025	[5]
GenImage++ CLIP-LoRA (OMAT)	Same backbone + LoRA, on-manifold adversarial fine-tune	Same SDv1.4 data + 6,000 SDv1.4-derived adversarial examples	2025	[5]

For the ResNet-50 the released code and the source paper disagree on input normalization, the former routing every detector through CLIP statistics and the latter specifying ImageNet statistics. We measured both and used ImageNet statistics, which performed better (85.3% versus 80.7% fake-accuracy on a 150-image FLUX.1 probe): a small instance of the under-specification problem Section 5 treats at length.

3.2. Generators and evaluation data

The five generators span the three generator families that account for essentially all current open-weight image synthesis, rather than sampling recent releases. BigGAN [8] is a class-conditional GAN, the family the first generation of detectors was built against. GLIDE [9] and SDXL [10] are U-Net diffusion models, early and modern, so the U-Net family contributes a within-family contrast of four years rather than a single point. FLUX.1 and SD3 replace the U-Net denoiser with a Diffusion Transformer [13], the architecture behind most 2024–2025 open releases. Grouping by family rather than release date is itself a result of this study (Section 4.1). Two families we do not cover are autoregressive token-based generators and face-specific manipulation pipelines, the second deliberately.

BigGAN and GLIDE come from the original GenImage benchmark [2], each evaluated against its own paired real split. FLUX.1, SD3 and SDXL come from GenImage++ [5], documented by its authors as test-only, which ships fake images only; since it has no matched reals, we use GenImage's BigGAN-subset real training split (about 162,000 ImageNet images, disjoint from the 6,000 used as BigGAN's own paired real evaluation split) as a shared real pool for all three. This asymmetry is forced by the dataset and we flag it rather than leave it implicit.

We use 6,000 real and 6,000 fake images for BigGAN, GLIDE, FLUX.1 and SD3, and the full 18,300–18,301 per class for SDXL, giving a worst-case 95% confidence-interval half-width of ±1.26 points at $n = 6{,}000$ against ±0.72 at $n = 18{,}300$; every gap we report is tens of points. One caveat: the dataset loader we inherited silently falls back to 100 images per class whenever the requested size exceeds either class pool, so a flag meaning "use all available data" had the opposite effect. Every number here comes from full-data reruns after that was found, and we mention it because the failure was silent and produced plausible-looking results.

3.3. Evaluation protocol

Each detector outputs one probability that an image is generated. We threshold at 0.5 for every accuracy figure, with no tuning. Because a fixed threshold can flatter or penalise a detector whose scores are shifted, we report threshold-free AUC alongside and treat calibration separately in Section 4.7.

We report real- and fake-accuracy separately, since a degenerate detector labelling everything "generated" would score 100% fake-accuracy. Real-accuracy stays between 86.0% and 98.9% across all 25 cells, so no detector approaches that failure mode; the converse case is visible rather than hidden, with the GenImage++ ResNet-50 on BigGAN at 98.5% real- and 0.4% fake-accuracy, close to labelling everything real.

3.4. Training/test leakage verification

Because the central claim here is that detectors fail to generalize, it matters that "fail to generalize" is not secretly "fail on data it never should have seen." The mirror image matters more, given how large the OMAT improvement in Section 4.5 is: "succeed" could equally be leakage. We checked training-data composition against primary sources rather than inferring it from the accuracy pattern the check exists to validate. UniversalFakeDetect and NPR both document ProGAN-only training in their own repository READMEs, with NPR's training command restricted to four ProGAN-generated LSUN categories and BigGAN appearing in that repository's folder layout only under a held-out test split. For the GenImage++ pair, the source paper states that "all detectors... were trained on the Stable Diffusion v1.4 (SDv1.4) subset of GenImage" and then fine-tuned on that same data augmented with $N = 6{,}000$ on-manifold adversarial examples (Zhou et al. [5], Section 6.1), with its Section 3.3 confirming those examples are themselves rendered from SDv1.4, and documents its own evaluation subsets as test-only. None of the five generators in our evaluation set appears in any of the five detectors' training data.

3.5. Relation to published numbers on these benchmarks

Before reporting new numbers we check the harness against figures the detectors' own authors published (Table 2). The NPR repository reports accuracies for the same ProGAN-4class checkpoint we evaluate, on

the AIGCDetectBenchmark assembly of BigGAN and GLIDE subsets. Those are assembled differently from GenImage's, so this is a consistency check rather than a reproduction; three of the four cells land within five points and the UniversalFakeDetect GLIDE cell within 0.1.

Table 2. Our overall accuracy against published figures for the same released checkpoints. The first four rows compare against the NPR repository's AIGCDetectBenchmark table, whose BigGAN and GLIDE subsets are assembled differently from GenImage's; the comparison is a consistency check, not a reproduction. The last row is a same-benchmark comparison, analysed in Section 5.

Detector	Generator	Published	Ours	Δ
NPR [4]	BigGAN	87.3	85.0	−2.3
NPR [4]	GLIDE	96.7	90.3	−6.4
UniversalFakeDetect [3]	BigGAN	95.1	90.2	−4.9
UniversalFakeDetect [3]	GLIDE	62.5	62.5	+0.1
GenImage++ CLIP [5]	BigGAN	62.7	61.7	−1.0

The last row is the one same-benchmark comparison, and it did not agree at first: a fake-only reading of our measurement was 9.6%, roughly six times below the published figure. Resolving that took a metric-definition correction plus a preprocessing change, and because the second moves any checkpoint on this benchmark we treat it as a result in its own right (Section 5).

4. Results

Table 3 gives fake- and real-accuracy for every detector and generator, and Table 4 gives threshold-free AUC and calibration error for the same cells, all under each detector's officially released preprocessing.

Table 3. Fake-accuracy / real-accuracy (%) at threshold 0.5. Bold marks each detector's own worst generator by fake-accuracy. Real-accuracy is reported alongside so the fake column cannot be read as evidence for a detector that simply predicts "generated" everywhere.

Detector	BigGAN	GLIDE	FLUX.1	SD3	SDXL
UniversalFakeDetect	83.0 / 97.4	27.4 / 97.7	7.0 / 96.8	11.7 / 96.8	38.5 / 96.8
NPR	84.1 / 86.0	94.3 / 86.2	97.7 / 88.1	90.1 / 88.1	54.6 / 87.7
GenImage++ ResNet-50	0.4 / 98.5	6.0 / 98.3	54.9 / 98.9	66.1 / 98.9	15.2 / 98.6
GenImage++ CLIP	9.6 / 96.5	69.1 / 96.0	86.5 / 97.4	88.6 / 97.4	77.0 / 97.4
GenImage++ CLIP-LoRA	99.7 / 94.8	99.9 / 94.9	96.5 / 96.0	98.3 / 96.0	99.2 / 94.7

Table 4. AUC (%) and expected calibration error (ECE, 15 bins, lower better) by detector and generator. AUC is threshold-free; ECE rises with failure severity rather than falling, the signature of confident error discussed in Section 4.7.

Detector (AUC / ECE)	BigGAN	GLIDE	FLUX.1	SD3	SDXL
UniversalFakeDetect	98.2 / .042	85.6 / .300	61.4 / .439	72.5 / .406	85.4 / .245
NPR	91.3 / .126	95.2 / .084	96.9 / .060	94.4 / .091	76.5 / .268
GenImage++ ResNet-50	51.4 / .503	60.3 / .469	86.2 / .212	92.8 / .153	68.8 / .415
GenImage++ CLIP	68.0 / .433	94.0 / .120	98.6 / .034	98.7 / .031	97.0 / .076
GenImage++ CLIP-LoRA	99.7 / .015	99.8 / .015	99.3 / .020	99.7 / .015	99.4 / .012

4.1. Detectors disagree on which generator is hardest

UniversalFakeDetect fails worst on FLUX.1 (7.0%), NPR on SDXL (54.6%), and both GenImage++ non-adversarial baselines on BigGAN (0.4% for ResNet-50, 9.6% for CLIP). CLIP-LoRA is lowest on FLUX.1 too, but at 96.5%, only 3.4 points below its own best, so "worst generator" is a weak label there.

The clearest illustration is the GenImage++ CLIP baseline, which catches 86–89% of FLUX.1 and SD3 fakes, the exact generators UniversalFakeDetect almost entirely misses, while catching under 10% of BigGAN, which UniversalFakeDetect and NPR both catch above 82% and CLIP-LoRA at 99.7%. If recency alone drove difficulty, no detector should recover on the oldest generator what it loses on the two newest. What predicts a weak spot is which generator family the training data did not cover, old or new. Nor is any generator here intrinsically easy or hard: every one spans at least 84 accuracy points across the five detectors, and BigGAN spans 99.3, holding both the lowest cell in the table (0.4%) and one of the highest (99.7%) on identical images. "This generator is hard to detect" is not well-formed without naming the detector.

4.2. The gaps are statistically significant, with one informative exception

All five detectors scored the identical fake-image set for each generator (6,000 images; 18,300 for SDXL), making the comparison paired and admitting McNemar's test [24]. For detectors A and B we count n_{01}, images A gets right and B does not, and n_{10}, the reverse, and run an exact two-sided binomial test on the discordant pairs. We restrict to fake images because the real-image sets are matched across detectors only for BigGAN and GLIDE (Section 8).

Table 5 gives the tests the argument rests on. The FLUX.1 contrast is decisive: NPR classifies 5,452 FLUX.1 fakes correctly that UniversalFakeDetect misses, against 8 the other way, $p < 10^{-300}$, opposite outcomes on the same images.

The informative exception is BigGAN, where UniversalFakeDetect and NPR score 82.95% and 84.12%, and it would be easy to call NPR better. The paired test disagrees: 835 images go one way and 905 the other, $p = 0.098$. Of 50 pairwise tests (ten detector pairs across five generators) this is the only one failing significance at $\alpha = 0.05$; the next weakest, at $p = 9.6 \times 10^{-3}$, would also fail under Bonferroni correction ($\alpha/50 \approx 0.001$), but every result our substantive claims rest on clears $p < 10^{-9}$, so family-wise correction changes no conclusion. The 1.2-point BigGAN gap should not be interpreted, which strengthens Section 4.1 rather than weakening it: BigGAN is where the two ProGAN-trained detectors behave alike, as a shared lineage predicts, while every cross-lineage comparison separates sharply.

Table 5. Paired McNemar tests on identical fake-image sets. n_{01} counts images the first detector gets right and the second does not; n_{10} the reverse. Exact two-sided binomial on discordant pairs. Only the first row is non-significant at $\alpha = 0.05$.

Generator	Comparison (A vs. B)	n_{01}	n_{10}	p
BigGAN	UniversalFakeDetect vs. NPR	835	905	0.098
BigGAN	CLIP vs. CLIP-LoRA	0	5406	$< 10^{-300}$
GLIDE	UniversalFakeDetect vs. NPR	97	4113	$< 10^{-300}$
FLUX.1	UniversalFakeDetect vs. NPR	8	5452	$< 10^{-300}$
SD3	NPR vs. CLIP	622	533	9.6×10^{-3}
SDXL	CLIP vs. CLIP-LoRA	74	4136	$< 10^{-300}$

4.3. Failure-profile shape tracks training data, its level tracks architecture

The two GenImage++ non-adversarial baselines separate those factors: they were trained on identical data (the GenImage SDv1.4 subset) and differ only in architecture, one a CLIP ViT-L/14 with a linear probe, the other a plain ResNet-50.

Their difficulty orderings are identical. Both rank the generators, hardest to easiest, BigGAN < GLIDE < SDXL < FLUX.1 < SD3. Spearman correlation between the two profiles is $\rho = +1.000$, and an exact permutation test over all $5! = 120$ orderings gives $p = 0.017$, the smallest value attainable at this sample size. No other pair matches orderings; the remaining nine correlations span $\rho = -0.800$ to $+0.700$, none significant at $n = 5$. Their levels are far apart: mean fake-accuracy is 28.5% for the ResNet-50 against 66.2% for the CLIP probe, a 37.6-point gap on identical training data and identical evaluation images.

Read together, these two observations point to a decomposition: the shape of a failure profile, which generators a detector struggles on relative to others, co-varies with the training distribution, while its

level co-varies with the representation. We state this as association rather than cause, since nothing here manipulates either factor. Shared training data alone does not produce a shared profile: UniversalFakeDetect and NPR are both ProGAN-trained yet anti-correlate ($\rho = -0.800$), differing in training subset (20 LSUN categories against 4) and in input representation, one a semantic embedding and the other a high-pass pixel residual.

This is a two-detector observation and we mark it as a hypothesis, not a finding. An exact-match ordering could not be tighter evidence at $n = 5$, but $n = 5$ is five, the permutation p of 0.017 is the floor of what this design can produce rather than a measure of effect size, and a shared quirk of the SDv1.4 subset could produce the same ordering twice for reasons unrelated to the general claim. Establishing it properly means training matched architectures over deliberately varied datasets, a training study outside our scope. Testing a prediction it makes is not.

4.4. Testing the prediction: ensembling by lineage, not by architecture

That decomposition makes a falsifiable prediction. If blind spots follow training data, combining two detectors of the same lineage should leave the shared blind spot in place however different their architectures are, while combining across lineages should cover it. The per-image scores already collected test it without retraining. We fuse pairs by averaging predicted probabilities at the same fixed 0.5 threshold used everywhere else, and report worst-generator fake-accuracy, since a blind spot is defined by the generator a combination handles least well. Real-image sets are shared across detectors only for BigGAN and GLIDE, so ensemble real-accuracy is reported for those two and omitted elsewhere.

Table 6. Mean-probability fusion at threshold 0.5. "Worst" is the lowest fake-accuracy across the five generators, the depth of the combination's blind spot. The first block holds the GenImage++ CLIP baseline fixed and varies only the partner. Real-accuracy is given for the two generators whose real splits are shared across detectors.

Combination	Lineage	Repr.	Mean fake	Worst fake	Real BG/GL
GenImage++ CLIP alone	–	–	66.2	9.6	96.5 / 96.1
+ GenImage++ ResNet-50	same	differs	38.0	0.3	99.2 / 99.0
+ UniversalFakeDetect	differs	same	52.8	39.7	99.3 / 99.1
+ NPR	differs	differs	84.6	62.9	92.7 / 93.1
NPR alone	–	–	84.2	54.6	86.0 / 86.2
+ UniversalFakeDetect	same	differs	83.1	53.2	91.9 / 92.6
GenImage++ CLIP-LoRA alone	–	–	98.7	96.5	94.8 / 94.9

The prediction holds, with a large margin. Table 6 holds the GenImage++ CLIP baseline fixed and varies only its partner. Adding the ResNet-50, which shares its training data and differs in architecture, takes worst-generator accuracy from 9.6% down to 0.3%: architectural variety within a lineage made the blind spot deeper, because both models are blind to BigGAN and averaging two confident wrong answers produces a confident wrong answer. Adding NPR, which shares no training data, lifts the same quantity to 62.9%, and a paired McNemar test separates the two combinations on every generator, the cross-lineage pair winning 4,284 BigGAN images to 3 at $p < 10^{-200}$ on all five. The pattern repeats on the other side: pairing NPR with UniversalFakeDetect, its ProGAN co-lineage partner, changes nothing (54.6% to 53.2%). Setting aside the CLIP/CLIP-LoRA pair, since a LoRA fine-tune of a checkpoint is not an independent detector, the six cross-lineage pairs average 52.5% worst-generator accuracy against 37.3% for the three same-lineage pairs.

One caution on what "lineage" is doing here. Detectors that differ in training data differ in other ways too: NPR and the GenImage++ CLIP baseline differ in architecture, input representation and training recipe as well as in data, so the contrast is between detector pedigrees rather than between training sets with everything else held fixed. The one genuinely controlled pair available, the two GenImage++ baselines, varies architecture alone, and that is the pair whose shared blind spot the fusion fails to cover. The result is consistent with lineage being the operative variable without isolating it.

Two qualifications keep this from being a recommendation to ensemble. Lineage diversity is necessary but not sufficient: NPR paired with the cross-lineage but very weak ResNet-50 lands at 47.0%, below NPR alone. And every ensemble here is beaten by a single detector, CLIP-LoRA, whose 96.5% worst-generator accuracy is more than 30 points above the best pair we construct, because averaging a strong detector's confident correct scores with a weak partner's confident wrong ones dilutes the strong one. The reading is about blind spots rather than leaderboards: when a combination must be built from detectors of comparable strength, choose them for different training lineages.

4.5. OMAT closes the gap, with a preprocessing caveat

The GenImage++ CLIP baseline and its OMAT-hardened LoRA counterpart share a backbone and evaluation images; the only difference is the adversarial fine-tuning step. Mean fake-accuracy rises from 66.2% to 98.7% (overall accuracy 81.6% to 97.0%), mean AUC from 91.3 to 99.6, and the spread of fake-accuracy across generators collapses from 79 points (9.6–88.6%) to 3.4 (96.5–99.9%). Every generator, including the three the baseline never saw, clears 96% fake-accuracy while real-accuracy stays within 94.7–96.0%, so the gain is not bought by shifting the operating point toward calling everything generated. It does not survive every preprocessing choice, however: swapping the official upsample for a native-resolution-preserving path on BigGAN collapses the OMAT model from 99.7% to 16.3% while improving the baseline (9.6% to 25.3%). Section 5.1 partly exonerates OMAT, since the same swap on GLIDE leaves it untouched and the BigGAN collapse is driven by a zero-padding border rather than by the change of interpolation.

4.6. Spectral deviation does not track detectability

A generator's spectral signature does not predict how detectable it is. On high-frequency deviation from real-photo statistics (mean log-magnitude FFT energy in the outer 20% of the frequency radius, 300-image samples), SD3 alone carries more high-frequency energy than real photos (+15.1%), against BigGAN (−283.5%), GLIDE (−234.3%), SDXL (−52.4%) and FLUX.1 (−31.6%). That ordering matches the detectability ordering in Table 3 under no detector: BigGAN and GLIDE deviate most yet sit at opposite ends of several rankings. With five generators we fit no correlation, but the two are plainly separate axes.

4.7. Detectors fail by being confident, not uncertain

The per-image scores show where the errors sit. For UniversalFakeDetect on FLUX.1 and SD3, the fake-image distribution does not sit near the 0.5 boundary; it collapses into the 0–0.1 bin, the region occupied by correctly classified real images. The failure mode is not borderline uncertainty but confident misclassification.

Reading that off a histogram is not enough, so we quantify it with expected calibration error [23] over 15 equal-width bins (Table 4) and show the underlying reliability curves in Figure 1. The numbers track the failures: UniversalFakeDetect's ECE is 0.042 on BigGAN, where it works, and 0.439 on FLUX.1, where it does not, meaning average confidence overstates average correctness by more than 43 points. A merely uncertain detector would push scores toward 0.5 and produce a small ECE alongside low accuracy; instead accuracy falls and calibration error rises together. The same holds for the CLIP baseline on BigGAN (ECE 0.433 at 9.6% fake-accuracy) and most starkly for the ResNet-50 on BigGAN, whose ECE of 0.503 accompanies 0.4% fake-accuracy. CLIP-LoRA is best-calibrated everywhere, never exceeding 0.020, and is the only panel whose curves track the diagonal across all five generators.

The diagrams add what the scalar hides: for every failing cell the largest bin is the rightmost one, so the mass of the distribution sits at maximum confidence regardless of whether the detector is right, and the curves separate vertically by generator rather than shifting left. A detector that fails confidently gives a downstream system no signal that it is outside its competence, and abstention and human-review routing both depend on low confidence tracking low reliability.

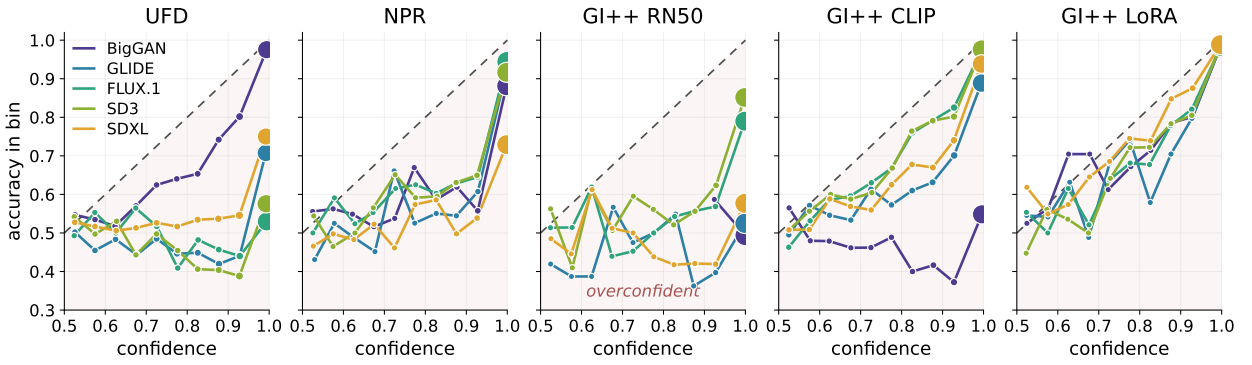


Fig. 1. Reliability diagrams, one panel per detector, five generators each. Confidence is $\max(p, 1-p)$ and accuracy is the fraction of images in the bin classified correctly, the quantities the reported ECE integrates. Marker area is proportional to the share of images in the bin, so the dominant mass near confidence 1 is visible rather than averaged away; bins holding fewer than 25 images are omitted, and the shaded region below the diagonal is overconfidence. A detector degrading into uncertainty would move left along the diagonal; these move downward instead, most visibly the GenImage++ CLIP baseline on BigGAN, whose largest bin sits at confidence 1.0 and 53% accuracy.

5. A preprocessing-driven reproducibility hazard

Resolving the disagreement left open in Section 3.5 turns up a sensitivity that applies to any cross-paper accuracy comparison in this literature. An implementation bug was ruled out first: reimplementing the source repository's released kornia-based preprocessing and running both versions over the same sixteen images through the same checkpoint gave near-identical scores, pointing at the preprocessing choice rather than a coding error.

GenImage's BigGAN subset ships fakes at 128×128, unlike every other generator in the benchmark (GLIDE ships at 256×256), which is the benchmark's documented convention rather than an artifact of our download. The released preprocessing resizes every input to 224×224 regardless of native size, so BigGAN's fakes are upsampled nearly twofold while others are downsampled or left near target. We added a second path preserving native pixel scale, pasting onto a black canvas rather than interpolating, and re-ran both checkpoints both ways (Table 7). Padding moves the CLIP baseline's overall accuracy to 61.66%, within one point of the reported 62.66%.

Two effects compound: the source paper's accuracy column is almost certainly overall accuracy across both classes while our first measurement was fake-only, a metric-definition mismatch on our end, and the resize direction is a real effect we measured rather than inferred. The same swap does the opposite to the OMAT checkpoint, dropping it from 99.7% to 16.3%, consistent with that paper's note that every fine-tuning strategy used a fixed resize-to-224 pipeline. We could not close the residual one point, because GenImage's official distribution is gated behind Baidu Yunpan and Google Drive with no scriptable, authentication-free access and the paper's cross-benchmark evaluation script is not among its released code.

5.1. Separating interpolation from border artifacts

An effect shown on one generator is an anecdote, so we repeated the test on GLIDE, the only other generator here whose native resolution is not 224×224. For the CLIP baseline it replicates in the same direction at smaller magnitude (fake-accuracy 69.1% to 76.9%, AUC 94.0 to 97.4, against 9.6% to 25.3% and AUC 68.0 to 90.4 on BigGAN), as the resolution gap predicts. For CLIP-LoRA the two diverge completely: on GLIDE it is unmoved (99.87% to 99.77%), on BigGAN it collapses (99.70% to 16.28%). Our native-resolution path center-crops images larger than 224 and zero-pads smaller ones, so for GLIDE's 256×256 inputs it is a pure center-crop with no border, while for BigGAN's 128×128 it produces a black frame covering 67.3% of the canvas. The BigGAN collapse is therefore confounded between the absence of interpolation and a large out-of-distribution border.

Table 7. Padding versus official upsampling on the two generators whose native resolution is not 224×224, full evaluation sets. The GLIDE padding condition is a pure center-crop; the BigGAN one introduces a black border over 67.3% of the canvas, so the two are not directly comparable.

Generator	Checkpoint / preprocessing	Fake	Real	AUC
BigGAN (128)	CLIP, upsample	9.60	96.48	67.97
BigGAN (128)	CLIP, padding	25.30	98.02	90.38
BigGAN (128)	CLIP-LoRA, upsample	99.70	94.82	99.70
BigGAN (128)	CLIP-LoRA, padding	16.28	93.12	69.13
GLIDE (256)	CLIP, upsample	69.13	96.05	94.03
GLIDE (256)	CLIP, padding	76.93	98.20	97.44
GLIDE (256)	CLIP-LoRA, upsample	99.87	94.87	99.82
GLIDE (256)	CLIP-LoRA, padding	99.77	92.82	99.88

We therefore ran a third condition on BigGAN: a nearest-neighbour upsample to 224×224, which removes smooth interpolation exactly as padding does but introduces no border. On matched 2,000-image-per-class samples the two causes separate cleanly. Dropping interpolation alone costs CLIP-LoRA 37.8 points of fake-accuracy (99.25% to 61.45%) while AUC falls only 1.7, from 99.63 to 97.90: an accuracy collapse with ranking intact is a threshold problem, the model still ordering generated above real while the score distribution shifts out from under the fixed cut. Adding the border is different damage, taking fake-accuracy to 5.30% and AUC to 60.36, near enough to chance that no threshold recovers it. This corrects Section 4.5: OMAT's hardening survives a change of interpolation kernel in ranking terms while losing threshold calibration, and fails outright only under a distortion with no counterpart in its training pipeline.

The CLIP baseline moves the opposite way on the same axis: under nearest-neighbour upsampling its AUC rises to 90.72 from 67.97, so for a detector never fine-tuned against a fixed pipeline, smooth interpolation of a 128×128 output is itself destroying signal. Two checkpoints trained on identical data therefore prefer opposite preprocessing, which makes a resize or crop mismatch a checkable explanation for an unexplained gap between published numbers, and makes any benchmark that fixes one convention a benchmark that quietly favours some detectors.

6. Compression: a robustness failure and a benchmark confound

Across all five generators, every generated image in these benchmarks is distributed as PNG and every real image as JPEG. Real and generated images therefore differ systematically in compression history, not only in provenance, and a detector sensitive to compression artifacts can score well by exploiting that. The bias is documented for GenImage by Grommelt et al. [25], who report that removing JPEG and image-dimension confounds shifts cross-generator performance by more than 11 points. It also matters on deployment terms, since images reaching a real detector have usually been re-encoded by whatever platform carried them.

One experiment answers both questions, run in three conditions on a fixed FLUX.1 sample of 2,000 images per class so that every condition is paired within a detector (Table 8). The one that isolates the confound is fakes-only, which re-encodes the generated images at JPEG quality 75 and leaves the reals untouched: it removes the format asymmetry and nothing else, so the drop from clean measures what that asymmetry was supplying. Both re-encodes every image, the deployment-style check and the closest available approximation to a compression-balanced benchmark.

NPR is the headline in both readings. It is the strongest detector on clean FLUX.1 by a wide margin (98.0% fake-accuracy, AUC 97.1), and compressing only the generated side takes its AUC to 39.13, not merely chance but below it: the detector then ranks real photographs as more likely generated than the generated images themselves. The inversion is what settles the interpretation. Destroying evidence drives AUC toward 50 and stops there, so a detector reading provenance alone cannot invert; a detector reading which side carries more high-frequency structure must invert, once the generated side becomes the more heavily compressed one. That is NPR's mechanism: it reads up-sampling artifacts in neighbouring-pixel

Table 8. FLUX.1, 2,000 images per class, the same sample in every condition. Fakes-only re-encodes generated images at JPEG q75 and leaves reals as shipped; both re-encodes every image. Fake-accuracy is shared between the two compressed conditions because the generated images are identical in both, so only AUC separates them. NPR falls below chance once generated images carry JPEG compression.

	Fake-acc (%)		AUC		
Detector	clean	JPEG q75	clean	fakes-only	both
UniversalFakeDetect	7.30	6.00	61.19	60.24	55.54
NPR	98.00	1.50	97.10	39.13	50.63
GenImage++ ResNet-50	69.45	58.00	91.27	87.02	88.48
GenImage++ CLIP	89.60	59.80	98.56	94.03	97.28
GenImage++ CLIP-LoRA	97.65	41.90	99.30	86.49	90.67

relationships, and JPEG quantisation attacks exactly the content those live in. The direction of NPR's evidence therefore follows the compression ordering of the two classes rather than their provenance, and that much is measured here rather than inferred. What the experiment does not settle is a quantity. Because re-encoding removes genuine generative traces at the same time, it bounds neither the share of the clean 98.0% that the format asymmetry supplied nor what would survive on a compression-balanced benchmark.

The other four detectors set the scale. Removing the format asymmetry costs the CLIP baseline 4.5 AUC points, the ResNet-50 4.3, CLIP-LoRA 12.8 and UniversalFakeDetect 1.0, against 58.0 for NPR, so the confound inflates an artifact-based detector by an order of magnitude more than a semantics-based one. Under full re-encoding their accuracy drops far exceed their AUC drops (the CLIP baseline loses 29.8 points of fake-accuracy and 1.3 of AUC), so the ranking still separates the classes while the fixed threshold no longer sits in the right place, which recalibration can fix. NPR's cannot, because no threshold helps at an AUC of 50.6, let alone 39.1: the detector a clean-data leaderboard recommends is the one that fails most completely on data that has passed through any normal image pipeline.

The confound is shared and symmetric, so it leaves the comparative results intact: every detector saw the same images, and the bias is a property of the real/fake pairing rather than of any detector, so it cannot produce the matched difficulty orderings of Section 4.3. Absolute accuracies should nonetheless be read as upper bounds, and repeating this study on a compression-balanced benchmark, such as the debiased GenImage variant released with [25], is the most important single follow-up.

7. Recommendations for practitioners

1. Choose a detector against the generator families you expect, not against a headline average: every generator here spans at least 84 accuracy points across five detectors.
2. If you combine detectors, combine across training lineages, and check first that one strong detector is not better than any pair you can build (Section 4.4).
3. Treat confidence as uninformative unless you have calibrated the detector on data resembling your deployment, since calibration error here rises with error rate.
4. Pin and publish the preprocessing: resize, crop, interpolation kernel and normalization statistics, alongside any accuracy you report.
5. Evaluate on re-encoded images before deployment. A detector reading high-frequency artifacts can lose all discriminative power under ordinary JPEG, and rank-based metrics will show this before accuracy does.
6. Do not use any of these detectors as sole evidence about a specific image or person.

For benchmark authors, one addition would have saved us days: a small, scriptable, authentication-free verification subset of a few hundred images with published checksums, plus the exact preprocessing function and metric definition behind the headline table.

8. Limitations

Detector diversity is narrower than "five detectors" suggests: three share a CLIP backbone family, so findings holding across all five are on firmer ground than those from that family alone. A genuinely independent family would widen it further. A diffusion-reconstruction method such as DIRE [15], whose signal is neither a CLIP embedding nor a pixel residual, is the most valuable single addition future work could make, and the obstacle is compute rather than availability. We obtained its released checkpoint and built a single-GPU harness for it, which we release alongside the rest of the code. DIRE scores an image by inverting it to a latent with DDIM and sampling back, roughly 40 diffusion forward passes per image, and that measured at 171 seconds per image on the 8 GB GPU used throughout. One hundred images per class on a single generator is 9.5 GPU-hours on that measurement, and matching the 6,000-per-class protocol the other five detectors run on, across five generators, would be near 2,800. We report the measurement rather than a partial sweep. The ensembling test likewise uses one fusion rule over pairs rather than larger sets, and its worst-generator statistic is a minimum over five generators and therefore sensitive to which five.

Cross-detector real-accuracy on FLUX.1, SD3 and SDXL is unpaired: because GenImage++ ships no reals, each run drew its own sample from the shared 162,000-image pool and those sets overlap only incidentally (7 of 6,000 for FLUX.1 and SD3, 200 of 18,300 for SDXL). The fake sets are identical across all five detectors for every generator, so fake-accuracy comparisons and every paired test are properly paired, and BigGAN and GLIDE are matched on both classes. Separately, whether the ImageNet images behind the GenImage++ checkpoints' training subset overlap our real pools is unresolved from released materials; it would not affect the fake-accuracy findings but could slightly inflate real-accuracy for those checkpoints, and their exact training composition rests on the source paper's reported methodology, which we could not check against training logs.

All detectors run at a fixed 0.5 threshold with no tuning, so the accuracy figures are out-of-the-box behaviour rather than each detector's best operating point, which is why AUC is reported alongside. On cost, the three CLIP detectors run at 7.4–7.6 images per second on our 8 GB GPU while NPR runs at 229–320, roughly 35 times faster. NPR is therefore both the cheapest detector here and the strongest on the two Diffusion Transformers under clean data, so the detector chosen on throughput is not the one chosen from headline accuracy, and neither choice survives Section 6.

9. Ethical and dual-use considerations

Documenting where detectors fail is dual-use: the table telling a platform which detector to avoid also tells an evader which generator to prefer. We think publication is still right, because the detectors and generators are already public and these failure modes are discoverable by anyone willing to run the same inference, the information asymmetry currently runs against defenders who deploy a single detector assuming a published number transfers, and our recommendations are defensive rather than evasive. Two cautions. None of these detectors should be sole evidence in a consequential decision about a person. And our results are measured on ImageNet-derived object and scene imagery; we make no claim about images of people, and face manipulation is out of scope entirely.

10. Conclusion

Three things follow from the measurements above, and each changes what a published accuracy number is worth.

A detector's weak spot is predicted by its training distribution, not by the generator's recency. That makes a single reported number close to uninformative about a deployment on a different generator family, and it makes the shape of a detector's failure profile the quantity worth reporting. The ensembling test in Section 4.4 is the practical consequence: combining for architectural variety within one training lineage left the shared blind spot in place and in one case deepened it, while combining across lineages covered it.

The failure mode is what makes that consequential. Calibration error here rises alongside error rate rather than falling, so a detector leaving its training distribution does not signal the fact through lower confidence, and the abstention and human-review policies built on confidence fail exactly where they are needed.

Both measurement hazards we report, the resize convention and the PNG-versus-JPEG asymmetry, point the same way: the absolute accuracies in this literature are conditional on conventions that are rarely stated. Cross-detector comparisons survive, since every detector here saw the same images, but absolute numbers should be read as upper bounds until benchmarks ship the verification subsets we recommend in Section 7.

Acknowledgements

Appendix A. Data and code availability

Detector checkpoints and evaluation datasets are the original authors' public releases, cited in Table 1 and Section 3. To make the preprocessing and metric definitions behind every number here checkable rather than described, the following are released together: the evaluation harness and the exact command line for each of the 25 detector/generator cells; the per-image prediction scores (image path, true label, predicted probability) for every cell, from which the paired tests, calibration computations and the ensembling analysis are computed; the preprocessing functions for the three resize conditions of Section 5 and the JPEG conditions of Section 6, including the normalization statistics used per detector; and the scripts that regenerate every results table and the figure from those score files. Everything listed above is available at https://github.com/MogSquad-FinalYear/BlindSpot, which also carries the per-image score files behind every table and the commands that produced them.

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